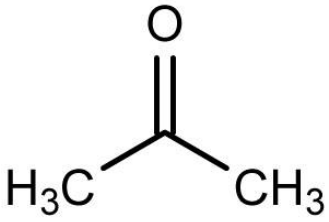
Organic chemistry lab

Infrared Spectra of Carbonyl Compounds

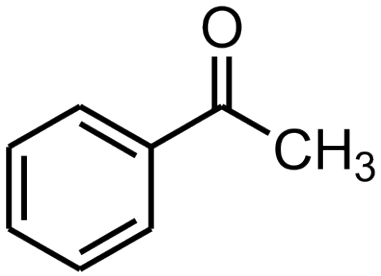
Theory:

Acetone



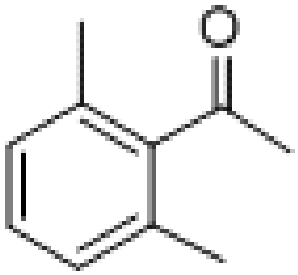
The organic compound acetone $(CH_3)_2CO$ is the simplest example of the ketones. Acetone is a colorless liquid that is miscible with water and is typically used as the solvent of choice in laboratory experiments. This compound's melting point is about 94˚C and its boiling point is about 56˚C.

Acetophenone



The organic compound acetophenone $C_6H_5C(O)CH_3$ is the simplest aromatic ketone. Like acetone it is also a colorless liquid. It's melting point is 19.5˚C and its boiling point is 202˚C.

2,6-dimethylacetophenone



The organic compound 2,6-dimethylacetophenone $C_{10}H_{12}O$. This compound has a boiling point around 110-112˚C.

Before you proceed further it may be beneficial to look over the material on infrared spectra in your lab manual.

This lab consists of 3 separate procedures that deal with 3 different compounds; therefore it would be helpful to read the following questions before beginning the experiments.

Questions:

How does the stretch frequency of acetophenone compare to that of acetone (longer or shorter)? Explain how your answer was determined. (Hint: draw contributing resonance structures)

Using the information gathered in this lab, is there a relationship between CO bond length and CO stretch frequency? Explain your answer.

Refer to the structures formed for acetophenone and 2,6 dimethylacetophenone, is one of these two molecules better suited for conjugation? If so why?

After comparing the stretch frequencies of 2,6-dimethylacetophenone, and acetophenone, does this tell you anything about the importance of conjugation in these systems?

Procedure A:

1. Open up the Spartan student program.
2. Click on the new button in the upper left corner.
3. Using the groups on the right under the organic tab form the compound acetone (C3H6O).
4. Go to the display tab at the top of the screen, then hit the spectra tab, and hit the draw experimental button. An IR spectrum should appear. Once this has occurred find the CO stretch frequency.
5. Once this frequency has been attained, form other compounds using other known groups and determine if there are any other frequencies that are similar to the CO frequency.

Procedure B:

1. Using the same program, form the compound acetophenone (C8H8O).
2. Go to the display tab, hit the spectra tab, and hit the draw experimental button to show the IR spectrum for this compound.
3. Determine the CO stretch frequency for this compound.
4. Next, go to the geometry tab, and under it click the measure distance button.
5. Then simply click on the two atoms in which the bond you are trying to measure connects and determine the CO bond length.

6. Compare with the CO bond length of acetone to determine if they are longer or shorter

Procedure C:

1. Using the same steps as in the previous procedure, form the compound 2,6-dimethylacetophenone (C10H12O)
2. Compare the geometry of 2,6-dimethylacetophenone to that of acetophenone (hint: consider orientation of phenyl ring and carbonyl group)
3. Next determine the CO stretch frequency for 2,6-dimethylacetophenone
4. How does this stretch frequency compare to that of acetophenone